Missing Data in Pandas

In Python, missing values refer to the absence of data in a particular variable or column of a dataset. Missing values are denoted by various representations, such as NaN (Not a Number), NA (Not Available), NULL, or simply an empty cell.There are several useful functions for detecting, removing, and replacing null values in Pandas DataFrame: 1.isnull() 2.notnull() 3.dropna() 4.fillna() 5.replace() 6.interpolate()

Checking for missing values using isnull()

In order to check null values in Pandas DataFrame, we use isnull() function this function return dataframe of Boolean values which are True for NaN values.

In [5]:
```
# importing pandas as pd
import pandas as pd
# importing numpy as np
import numpy as np
```

In [6]:
```
# dictionary of lists
dict = {'First Score':[100, 90, np.nan, 95],
        'Second Score': [30, 45, 56, np.nan],
        'Third Score':[np.nan, 40, 80, 98]}
```

In [7]:
```
# creating a dataframe from list
df = pd.DataFrame(dict)
```

In [8]:
```
# using isnull() function
df.isnull()
```

Out[8]:

	First Score	Second Score	Third Score
0	False	False	True
1	False	False	False
2	True	False	False
3	False	True	False

In [13]:
```
# making data frame from csv file
data = pd.read_csv("employees.csv")
data.head()
```

Out[13]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	NaN
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services

In [16]:
```
data.shape
```

Out[16]:
```
(1000, 8)
```

In [17]:
```
data.isnull()
```

Out[17]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	True
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...
995	False	True	False	False	False	False	False	False
996	False	False	False	False	False	False	False	False
997	False	False	False	False	False	False	False	False
998	False	False	False	False	False	False	False	False
999	False	False	False	False	False	False	False	False

1000 rows × 8 columns

In [20]:
```
data.isnull().sum()
```

Out[20]:
```
First Name            67
Gender               145
Start Date             0
Last Login Time        0
Salary                 0
Bonus %                0
Senior Management     67
Team                  43
dtype: int64
```

In [21]:
```
data.isnull().sum().sum()
```

Out[21]:
```
322
```

Filling Null Values

In [32]:
```
df1 =data.fillna(value=0)
data
```

Out[32]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	NaN
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	NaN	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [33]:
```
df1 =data.fillna(value=0)
df1
```

Out[33]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	0
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	0	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [34]:
```
df1.isnull().sum().sum()
```

Out[34]: 0

Filling null values with the previous ones

In [36]:
```
df2 = data.fillna(method = 'pad')
df2
```

Out[36]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	Marketing
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	Male	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [37]:
```
df2.isnull().sum()
```

Out[37]:
```
First Name           0
Gender               0
Start Date           0
Last Login Time      0
Salary               0
Bonus %              0
Senior Management    0
Team                 0
dtype: int64
```

Filling null value with the next ones

In [38]:
```
df3 = data.fillna(method = 'bfill')
df3
```

Out[38]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	Finance
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	Male	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [39]:
```
df4 = data.fillna(method = 'pad', axis = 1)
df4
```

Out[39]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.17	True	True
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.34	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	Henry	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [40]:
```
df5 = data.fillna(method = 'bfill', axis = 1)
df5
```

Out[40]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Tear
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketin
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.17	True	Nal
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Financ
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.34	True	Financ
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Clier Service
...	...	...	...	...	...	...	...	.
995	Henry	11/23/2014	11/23/2014	6:09 AM	132483	16.655	False	Distributio
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Financ
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Produc
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Busines Developmer
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sale

1000 rows × 8 columns

In [41]:
```
data
```

Out[41]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	NaN
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	NaN	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [42]:
```
#Filling different values in Null in different columns
df6 = data.fillna({'Team':'abcd',
                   'Senior Management': 'efgh'})
df6
```

Out[42]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	abcd
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	NaN	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [44]:
```
#Filling Null Values with the mean of a column
df7 = data.fillna(value= data['Salary'].mean())
df7
```

Out[44]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	90662.181
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	90662.181	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [45]:
```
df7 = data.fillna(value= data['Salary'].max())
df7
```

Out[45]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	149908
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	149908	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

df7 = data.fillna(value= data['Salary'].min()) df7

In [57]:
```
# Printing the first 10 to 24 rows of
# the data frame for visualization
data[23:40]
```

Out[57]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
23	NaN	Male	6/14/2012	4:19 PM	125792	5.042	NaN	NaN
24	John	Male	7/1/1992	10:08 PM	97950	13.873	False	Client Services
25	NaN	Male	10/8/2012	1:12 AM	37076	18.576	NaN	Client Services
26	Craig	Male	2/27/2000	7:45 AM	37598	7.757	True	Marketing
27	Scott	NaN	7/11/1991	6:58 PM	122367	5.218	False	Legal
28	Terry	Male	11/27/1981	6:30 PM	124008	13.464	True	Client Services
29	Benjamin	Male	1/26/2005	10:06 PM	79529	7.008	True	Legal
30	Christina	Female	8/6/2002	1:19 PM	118780	9.096	True	Engineering
31	Joyce	NaN	2/20/2005	2:40 PM	88657	12.752	False	Product
32	NaN	Male	8/21/1998	2:27 PM	122340	6.417	NaN	NaN
33	Jean	Female	12/18/1993	9:07 AM	119082	16.180	False	Business Development
34	Jerry	Male	1/10/2004	12:56 PM	95734	19.096	False	Client Services
35	Theresa	Female	10/10/2006	1:12 AM	85182	16.675	False	Sales
36	Rachel	Female	2/16/2009	8:47 PM	142032	12.599	False	Business Development
37	Linda	Female	10/19/1981	8:49 PM	57427	9.557	True	Client Services
38	Stephanie	Female	9/13/1986	1:52 AM	36844	5.574	True	Business Development
39	NaN	Male	1/29/2016	2:33 AM	122173	7.797	NaN	Client Services

Dropna()

In [58]:
```
df8 = data[23:40].dropna()
df8
```

Out[58]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
24	John	Male	7/1/1992	10:08 PM	97950	13.873	False	Client Services
26	Craig	Male	2/27/2000	7:45 AM	37598	7.757	True	Marketing
28	Terry	Male	11/27/1981	6:30 PM	124008	13.464	True	Client Services
29	Benjamin	Male	1/26/2005	10:06 PM	79529	7.008	True	Legal
30	Christina	Female	8/6/2002	1:19 PM	118780	9.096	True	Engineering
33	Jean	Female	12/18/1993	9:07 AM	119082	16.180	False	Business Development
34	Jerry	Male	1/10/2004	12:56 PM	95734	19.096	False	Client Services
35	Theresa	Female	10/10/2006	1:12 AM	85182	16.675	False	Sales
36	Rachel	Female	2/16/2009	8:47 PM	142032	12.599	False	Business Development
37	Linda	Female	10/19/1981	8:49 PM	57427	9.557	True	Client Services
38	Stephanie	Female	9/13/1986	1:52 AM	36844	5.574	True	Business Development

In [59]:

```
#all value
df9 = data[23:40].dropna(how='all')
df9
```

Out[59]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
23	NaN	Male	6/14/2012	4:19 PM	125792	5.042	NaN	NaN
24	John	Male	7/1/1992	10:08 PM	97950	13.873	False	Client Services
25	NaN	Male	10/8/2012	1:12 AM	37076	18.576	NaN	Client Services
26	Craig	Male	2/27/2000	7:45 AM	37598	7.757	True	Marketing
27	Scott	NaN	7/11/1991	6:58 PM	122367	5.218	False	Legal
28	Terry	Male	11/27/1981	6:30 PM	124008	13.464	True	Client Services
29	Benjamin	Male	1/26/2005	10:06 PM	79529	7.008	True	Legal
30	Christina	Female	8/6/2002	1:19 PM	118780	9.096	True	Engineering
31	Joyce	NaN	2/20/2005	2:40 PM	88657	12.752	False	Product
32	NaN	Male	8/21/1998	2:27 PM	122340	6.417	NaN	NaN
33	Jean	Female	12/18/1993	9:07 AM	119082	16.180	False	Business Development
34	Jerry	Male	1/10/2004	12:56 PM	95734	19.096	False	Client Services
35	Theresa	Female	10/10/2006	1:12 AM	85182	16.675	False	Sales
36	Rachel	Female	2/16/2009	8:47 PM	142032	12.599	False	Business Development
37	Linda	Female	10/19/1981	8:49 PM	57427	9.557	True	Client Services
38	Stephanie	Female	9/13/1986	1:52 AM	36844	5.574	True	Business Development
39	NaN	Male	1/29/2016	2:33 AM	122173	7.797	NaN	Client Services

In [60]:
```
#any value
df9 = data[23:40].dropna(how='any')
df9
```

Out[60]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
24	John	Male	7/1/1992	10:08 PM	97950	13.873	False	Client Services
26	Craig	Male	2/27/2000	7:45 AM	37598	7.757	True	Marketing
28	Terry	Male	11/27/1981	6:30 PM	124008	13.464	True	Client Services
29	Benjamin	Male	1/26/2005	10:06 PM	79529	7.008	True	Legal
30	Christina	Female	8/6/2002	1:19 PM	118780	9.096	True	Engineering
33	Jean	Female	12/18/1993	9:07 AM	119082	16.180	False	Business Development
34	Jerry	Male	1/10/2004	12:56 PM	95734	19.096	False	Client Services
35	Theresa	Female	10/10/2006	1:12 AM	85182	16.675	False	Sales
36	Rachel	Female	2/16/2009	8:47 PM	142032	12.599	False	Business Development
37	Linda	Female	10/19/1981	8:49 PM	57427	9.557	True	Client Services
38	Stephanie	Female	9/13/1986	1:52 AM	36844	5.574	True	Business Development

replace()

In [63]:
```
df10 = data.replace(to_replace=np.nan, value=1234)
df10
```

Out[63]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	1234
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	1234	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

In [66]:
```
df10 = data[23:40].replace(to_replace=np.nan, value=1234)
df10
```

Out[66]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
23	1234	Male	6/14/2012	4:19 PM	125792	5.042	1234	1234
24	John	Male	7/1/1992	10:08 PM	97950	13.873	False	Client Services
25	1234	Male	10/8/2012	1:12 AM	37076	18.576	1234	Client Services
26	Craig	Male	2/27/2000	7:45 AM	37598	7.757	True	Marketing
27	Scott	1234	7/11/1991	6:58 PM	122367	5.218	False	Legal
28	Terry	Male	11/27/1981	6:30 PM	124008	13.464	True	Client Services
29	Benjamin	Male	1/26/2005	10:06 PM	79529	7.008	True	Legal
30	Christina	Female	8/6/2002	1:19 PM	118780	9.096	True	Engineering
31	Joyce	1234	2/20/2005	2:40 PM	88657	12.752	False	Product
32	1234	Male	8/21/1998	2:27 PM	122340	6.417	1234	1234
33	Jean	Female	12/18/1993	9:07 AM	119082	16.180	False	Business Development
34	Jerry	Male	1/10/2004	12:56 PM	95734	19.096	False	Client Services
35	Theresa	Female	10/10/2006	1:12 AM	85182	16.675	False	Sales
36	Rachel	Female	2/16/2009	8:47 PM	142032	12.599	False	Business Development
37	Linda	Female	10/19/1981	8:49 PM	57427	9.557	True	Client Services
38	Stephanie	Female	9/13/1986	1:52 AM	36844	5.574	True	Business Development
39	1234	Male	1/29/2016	2:33 AM	122173	7.797	1234	Client Services

interpolate()

In [70]:
```
data['Salary'] = data ['Salary'].interpolate(method = 'linear')
data
```

Out[70]:

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
0	Douglas	Male	8/6/1993	12:42 PM	97308	6.945	True	Marketing
1	Thomas	Male	3/31/1996	6:53 AM	61933	4.170	True	NaN
2	Maria	Female	4/23/1993	11:17 AM	130590	11.858	False	Finance
3	Jerry	Male	3/4/2005	1:00 PM	138705	9.340	True	Finance
4	Larry	Male	1/24/1998	4:47 PM	101004	1.389	True	Client Services
...	...	...	...	...	...	...	...	...
995	Henry	NaN	11/23/2014	6:09 AM	132483	16.655	False	Distribution
996	Phillip	Male	1/31/1984	6:30 AM	42392	19.675	False	Finance
997	Russell	Male	5/20/2013	12:39 PM	96914	1.421	False	Product
998	Larry	Male	4/20/2013	4:45 PM	60500	11.985	False	Business Development
999	Albert	Male	5/15/2012	6:24 PM	129949	10.169	True	Sales

1000 rows × 8 columns

Interpolation in pandas is a method to fill missing values based on existing data patterns (like linear, time-based, or index-based)

In [3]:
```
import pandas as pd

# Sample DataFrame with missing values
data = {
    "Day": [1, 2, 3, 4, 5, 6, 7],
    "Math": [50, None, 70, None, 90, None, 100],
    "Science": [30, 40, None, 60, None, 80, None]
}

df = pd.DataFrame(data)
"Original DataFrame:"
df
```

Out[3]:

	Day	Math	Science
0	1	50.0	30.0
1	2	NaN	40.0
2	3	70.0	NaN
3	4	NaN	60.0
4	5	90.0	NaN
5	6	NaN	80.0
6	7	100.0	NaN

In [4]:
```
# Interpolating missing values for all numeric columns
df_interpolated = df.interpolate()

"\nAfter Interpolation:"
df_interpolated
```

Out[4]:

	Day	Math	Science
0	1	50.0	30.0
1	2	60.0	40.0
2	3	70.0	50.0
3	4	80.0	60.0
4	5	90.0	70.0
5	6	95.0	80.0
6	7	100.0	80.0

Lab Task

In [ ]:
```
Your task is to apply the following operations on your dataset
```

isnull() notnull() dropna() fillna() replace() interpolate()

Send me the task on my email. Make sure everyone uses their own dataset

Email: aqsakhann913@gmail.com

In [ ]: